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客户 Customer	通用技术文档
项目 Project	通用技术文档
主题 Title	ABS9/TCS9/ESP9 线束安装说明
内容概要 Summary	本文主要给出了ABS9/TCS9/ESP9 线束特性描述和安装说明 在任何有异议的情况下，本文的最终解释以英文版本文件Y265K30179为准。

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ESP9 通用 B1 样品电路图

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1. 产品和应用目标 Product and Application Purpose

本文档的目的是，以博世的专业意见为出发点，遵照客户的信息和要求为客户提供合适的、正确的线束安装说明。请注意，博世对此文中的推荐信息不承担任何必然的责任，确保线束密封性和安装正确性的责任完全取决于客户。

Purpose of this document is to set out for the customer information and requirements for what, according to our opinion, may be considered a proper and correct wiring harness installation. Please note that Bosch does not assume any liability for the information/recommendations set out herein and that responsibility for securing the tightness and the proper installation of the wiring harness solely rests with the customer.

ABS9/TCS9/ESP9 线束主要用于连接 ABS9/TCS9/ESP9 部件。对轮速传感器的布线安放和线缆之间的电磁兼容性要特别注意。线束连接图是博世对车辆线束接口的最基础要求。任何与线束连接图有出入部分均需要与博世进行协商。

The ABS9/TCS9/ESP9 wiring harness has the task of electrical connection of the ABS9/TCS9/ESP9 components. Special attention should be given to the arrangements of the wheel speed sensors and the recommendation to EMC. The wiring diagram is Bosch’s minimum request of the vehicle wiring interface. Any deviations from this wiring diagram should be clearly communicated to Bosch.

每一种车型均有其相应的线束连接图，图中包含了完整的针脚分布信息以及连接器类型。

A wiring diagram is supplied for each vehicle type with complete pin assignments. Information about connector type is also included.

2. 环境条件 Environmental Conditions

线束必须达到或超过液压单元客户技术文档 Y 265 K40 200 最新版本规定的要求。

The wiring harness should meet/exceed the minimum requirements which are specified in the Hydraulic Unit Technical Customer Document Y 265 K40 200 latest version.

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3. 线缆横截面 Wire Cross Section

3.1 关键线缆 Critical Wires

线缆截面积和标称线缆长度推荐参考文献 Y 265 B32 347/E。当确定线缆规格时，请确保线缆阻值满足第20页中的要求。如果各线缆的阻值不能满足要求，博世需要考虑电压降情况。

Recommendations for wire cross section and nominal vehicle wire length are included in Y 265 B32 347/E. When determining the wire gauges, make sure that the wire resistance meets the specification on page 20. If the respective resistance cannot be maintained, a separate voltage drop calculation is necessary by Bosch.

3.2 非关键线缆（低电流设计）Non Critical Wires (low current draw)

非关键线缆存在电压降，客户应当根据机械应力要求以及连接器、接线端等方面的要求来选择线缆(0.35 mm^2....0.8 mm^2).

The non critical wires in regards to voltage drop can be selected by the customer according to their own mechanical stress requirements and to the requirements of the connector and its terminals (0.35 mm^2....0.8 mm^2).

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4. 轮速传感器布线要求 Wheel Speed Sensor Wiring

轮速传感器线缆即接头K1到接头K12、K14、K16、K18的双绞线。这些双绞线要求每隔 40-58mm 绞绕一次。由于从轮速传感器到ECU的连接要求较高,建议选择高质量的线缆供应商，以此来满足20页中提到的漏电流要求(绝缘电阻)。

From connector K1 to connector K12, 14, 16, 18. These two wires must be twisted with a length l = 40 - 58 mm per one twist. Due to the high requirements of this connection (speed sensor - ECU) it is proposed to use only high quality material suppliers. This assures that the leakage requirements/isolation resistance described on page 20 will be met.

如果轮速传感器线缆存在强信号干扰等电磁兼容性问题,建议使用双绞线加屏蔽。屏蔽单独接地。其他解决方法也可能达到相应效果，但须得到博世认可。

If there are problems with the EMC on the wheel speed sensor wire (e.g. high interference), the application of a twisted wire with an additional shield is recommended. The shield is separately connected to ground. Deviations from this recommendation are still possible, but should be agreed by Bosch.

4.1 CAN布线要求 CAN Wiring

对于 CAN 线，建议使用每隔 40-58 毫米绞绕一次的双绞线。如果轮速传感器线缆存在电磁兼容性问题，建议使用双绞线加屏蔽。屏蔽单独接地。

For the CAN wiring it is recommended to use also a twisted pair with a length l = 40 - 58 mm per one twist. If there are any problems with the EMC the use of a twisted wire with an additional shield is recommended. The shield is separately connected to ground.

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5. 布线 Wiring Layout

5.1

ABS9/TCS9/ESP9 线束要求绕开发动机控制模块以及其他电子模块。应尽量使ABS9/TCS9/ESP9 线束与其它导线保持足够远的距离。如果要计划与其它线束一起采用常规布线，建议联系博世。

The ABS9/TCS9/ESP9 wiring harness should be routed away from the engine control module and other electronic modules. All attempts should be made to maintain as much distance as possible between the ABS9/TCS9/ESP9 harness and other conductive wiring. If a common layout with other wiring harness is planned, it is recommended to contact Bosch.

5.2

电源线1和25必须直接连到电源上，以尽量减少由其他电路或电路模块引起的电压损耗。不允许直接使用一根线作为公共电源线并将ABS9/TCS9/ESP9 和其他高电流负载都接到这根线上。

The power leads 1 and 25 must be connected directly to the battery itself to minimize a voltage drop created by other electric or electronic devices. It is not allowed to create one common power line and splice this for ABS9/TCS9/ESP9 and other high current consumers.

如果需要采用其他连接方式，必须考虑电压降问题。博世可以完成这方面工作，但需要提供线束的信息，包括线缆截面积、长度等。

If a different connection becomes necessary a separate voltage drop calculation should be made. This can also be done by Bosch when all harness information like cross section and cable length is available.

5.3

引线13 (回流泵地线)和引线38 (ECU 和电磁阀地线)一定要一起接到同一处车身搭铁或电源负极接线柱上。与电源线一样，地线也需要单独的接地接线柱。同样也不允许把几个负载用一根公共地线接地。这会使 ABS9/TCS9/ESP9 系统的地

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电位发生漂移。

The connection of wire 13 (ground wire recirculation pump) and wire 38 (ground wire ECU and solenoid valves) must be connected together to one common ground stud at the car body or battery terminal. Similar to the power supply also the ground must have its own ground stud. It is also not allowed to splice different consumers with one common ground wire. This could create an additional ground shift for the ABS9/TCS9/ESP9 system.

5.4

5.4.1

轮速传感器线缆和制动灯开关信号线不允许与高电压引线(如点火线)并行。如果交叉的话， 必须保证距离在10厘米以上。避免和喷油器、PWM 线和高压点火/电源线接近，即使是很短的一段距离也应避免。为了能在包括系统故障的各种情况下进行监测，应当有额外的屏蔽措施。在任何情况下，其他内部设备都需要进行偏移电磁场测试。

The wheel speed sensor wiring and the brake light switch wire may not be guided in parallel to high voltage wiring (e.g. ignition cable). In case of crossing a minimum distance of d > 10 cm is required. Avoid proximity to fuel injectors, PWM lines or high power ignition/battery wiring, even for short distance. In order to make proper detection, even in case of failure, extra screening measures are necessary. In any case, a stray electrical field test is necessary for other internal devices.

5.4.2

ESP9 系统安装了转向角传感器和横摆角速度传感器。使用博世 ESP9 的传感器，电源线必须在 ECU 10A 保险丝之后直接引出到各传感器。10A 保险丝安装在点火线和 K1/32 之间。这样的话，这些部件(转向角传感器和横摆角速度传感器)可以分担 ECU 的负载过流保护。如果采用其他电源连接方式，当过载保护出问题时，博世将不承担相应责任。如果使用 CAN 线横摆角速度传感器和 CAN 线转向角传感器，则传感器可以直接接到底盘地。

The ESP9 system is using a steering angle sensor and a yaw rate sensor. By using the Bosch sensors for ESP9 it is required to make the connection for the power supply for both components directly after the 10 A fuse for the ECU. This 10 A fuse is installed in the wire between ignition and K1/32. In this case both components (steering angle sensor and yaw rate sensor) can participate from the load dump protection of the ECU. If another power supply is chosen

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Bosch does not assume any responsibilities in case of any failure during a load dump. If a CAN yaw rate sensor and CAN steering angle sensor are used they can wired directly to chassis ground.

5.4.3

ESP9 接收和发送来自其他 ECU (如 Motronic 数字发动机管理系统/发动机控制器) 的 CAN 信号。为避免由于电压不同和开关的闭合引起的故障，建议不同的 ECU 使用相同的电源和接地。

The ESP9 transmits/receives different CAN messages from other ECU's (e.g. Motronic/engine controller...). To prevent failures due to different voltage levels and switch ON/OFF behavior it is recommended to use the same power supply and ground level for the different ECU's.

5.5

5.5.1

在引线15(点火线)和 ESP9 告警灯之间不需要安装保险丝，因为一旦保险丝熔断的话，ESP9 的故障无法检测到。如果有特例(例如，其他灯和保险丝连在一起的话，如果保险丝断了，可以直接显示到驾驶员那里)，应征得博世的同意。

No fuse must be installed between terminal 15 (ignition) and ESP9 warning lamp, because if the fuse is blown an ESP9 failure cannot be detected. Deviations require the approval of Bosch, (e.g. other lamps are fused together so that a blown fuse can be indicated immediately to the driver).

5.5.2

博世建议以下引线保险丝：Bosch recommends fuses for the following wires:

引线1 (电源到 ECU) 40A no. 1 (between battery and ECU) 40 A
引线25 (电源到 ECU) 25A no. 25 (between battery and ECU) 25 A
引线7 (电源到 ECU) 5A no.7 (between battery and ECU) 5A
引线32 (点火器到 ECU) 10A no. 32 (between ignition key and ECU) 10 A

这些保险丝必须满足失效模式分析(FMEA)的要求，以此来避免由于短路引起的线束过热。引线1上保险丝的值同样取决于环境条件温度以及 ABS9 作用时主缸的压力。

These fuses are necessary to fulfill the requirements of failure mode effect analysis (FMEA) to avoid overheating of the wiring harness in case of a short

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circuit. The value of the fuse in wire 1 depends also on the environmental conditions of temperature and applied pressure at the master cylinder during an ABS9 actuation.

保险丝的初始值由博世设定，在征得博世同意以后可以进行更改，如保险丝的不同熔断特点。使用熔线则是不允许的，因为该器件的熔断特点有很大不同。

This value is a first assumption of Bosch and can be changed after approved by Bosch, e.g. different characteristics of the fuse blow characteristic. The use of fusible links is not allowed, because the blow characteristic of this device is very different.

在最坏的情况下，即 ABS9 制动和 TCS9 作用时，电流的测量结果将如第18，19页中的图所示。在多数情况下(20-50 Hz)下泵马达的电流将被调制，因此无法得到稳定的电流。同时，在多数情况下(4 kHz)一部分电磁阀(压力保持和回路控制阀)也将被调制，因此有可能出现电压尖峰。

Current measurements during a worst case ABS9 braking and a typical TCS9 actuation in regards to the current is shown at page 18 and 19. Due to the fact that the current for the pump motor is modulated in most cases (20-50Hz) a steady state current is not possible. Also a part of the solenoid valves (pressure holding- and pilot valves) are modulated in most cases (4 kHz) and therefore spikes at the voltage level are possible.

给 ECU(K1/32) ，泵马达(K1/1)和电磁阀(K1/25)供电的电源电压必须一直保持在 10V 以上，以避免电压不足导致系统关闭。

The voltage level at the terminal power supply (K1/32) for ECU, pump motor (K1/1) and solenoid valves (K1/25) must be in all cases > 10V to prevent any undervoltage shutoff in any situations.

5.6 连接器 Connectors

由于系统的连接器有不同的供应商，在各个部件的图纸中附有连接器的信息。

Due to the different connector supplier for the system the additional connector information is shown in each component offer drawing.

附 ECU 连接器的信息：这是一个38针脚的连接器，有4根电源线 (配单独的密封圈)和34根信号线 (可以共用一个整体密封圈或各自独立密封)。连接器要满足

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系统要求、引脚分布以及信号要求。

Additional information for the ECU-connector:

This is a 38-pin connector with 4 power terminals with single sealing and 34 signal terminals with a common family seal or also single sealing. The connector depends on the systems and is defined due to the different pin assignment and signal requirements.

不同的线径有不同的密封特性，根据这些特性制定出来的连接器密封要求必须满足，并由线束供应商随后根据整车厂的密封要求进行检查。

The special connector requirements in regards to tightness according to the sealing behavior of different wire diameters must be kept and checked by the wiring harness supplier according to the OEM's tightness requirements.

5.6.1 线束固定 Fixing of the Wiring Harness

考虑到机械应力和振动，建议将线束固定在38针脚连接器后至少200到400毫米处。博世建议直接把线束固定在可运动部件如液压单元上，这样可以避免38针脚连接器上的接线端损坏。

Due to the mechanical stress and vibration of the wiring harness the wiring harness must be fixed at least 200mm - 400mm after the 38-pin connector. Bosch recommends to fix the wiring harness directly at the hydraulic unit in a way that the wiring harness is fixed at the moving element, e.g. hydraulic unit. With this solution it is prevented that the terminals in the 38-pin connector are not damaged.

5.7 告警灯 Warning Lamp

为了满足 ECE13 对告警灯激活的要求，必须强制配备主动式告警灯。ABS、TCS、ESP的告警灯可以通过硬线或通过 CAN 线连到系统上。这取决于客户的布线和激活告警灯的要求。主动式告警灯：当地线断了或松了时，ESP 将被关闭，相应地告警灯将激活。

To fulfill the requirements of ECE 13 in regards to the actuation of the warning lamp an active warning lamp is mandatory. The warning lamps (ABS, TCS, ESP) can be hard wired or also actuated via CAN. This depends on the customer layout of the wiring and actuating of the warning lamps.

- active warning lamp: in case of broken wire or loss of ground wire ESP

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switching OFF and therefore the warning lamps can be actuated.

5.8 对电磁兼容性的建议 EMC Recommendations

针对不同的部件安装位置和布线，下面是几点常规建议：
-轮速传感器沿车左边布线，电源线沿车右边布线；
-不要把轮速传感器线和高压线(点火线)并行布线，应保持最小10厘米的距离；
-连接 ECU 到电源的线应该尽可能的短；
- ECU 接车底盘的地线应该尽可能短或直接连到电源地；
-避免绕圈过长；
-避免系统中不使用的线缆==>> 会对 ABS9/TCS9/ESP9 线束造成干扰；
-使诊断连接器尽可能靠近 ABS9/TCS9/ESP9；
-诊断线要尽可能短并尽量满足客户要求；
-如果线束布线改变，要重新进行 EMC 测试；
-一般来说高速线缆如 CAN 线是关键线缆 (保持一定电阻值，如120欧姆)；
-如果需要其他的支持，请联系博世公司。

These are general recommendations depending on the different locations of the components and wiring in the car.
- wheel speed sensor wires at the left side of the car and power cable routing at the right side
- do not route the wheel speed sensor wires in parallel to high tension wires (ignition cables), a minimum distance of more than 10 cm is necessary
- the wire from ECU to battery should be as short as possible
- ECU ground as short as possible fixed to chassis or directly to battery ground
- avoid loops in the routing
- avoid wires which are not in use for the system
  ==>>interference for the ESP9 wiring harness
- place the diagnostic connector as close as possible to the ESP9 unit
- diagnostic wires as short as possible and closed for the customer specific requirements
- if the wiring harness routing is changed the EMC-tests should be performed again
- wires with high frequency e.g. CAN wires are critical in general (keep resistance e.g. 120 Ohm)
- if additional support is needed, please contact Bosch

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6. 附密封要求和ECU通风要求 Additional Sealing Requirements and Ventilation of the ECU

通风 Ventilation

ABS9/TCS9/ESP9 的 ECU 是通过 DAE(一种压力平衡装置，即 Gore-tex 防水多孔膜)来通风的。通过 DAE 实现的 ECU 通风并不能阻止水分通过开放或损坏的线缆渗入线束并进入连接器，这样系统将因连接器腐蚀而发生故障。为了避免上述情况，所有的线束都必须达到本章所列的要求。

The ABS9/TCS9/ESP9 ECU is ventilated via a DAE (pressure balance equipment = goretex membrane) The ventilation of the ECU by this DAE does not prevent any water penetration through open or damaged wires into the wiring harness and through the wires into the connector so that the system can fail due to corrosion in the connector. Therefore all wiring harness requirements must be kept as defined in chapter 6.

地线和接合处的密封要求 Sealing Requirements for Ground Wires and Splices

当有水溅到 ABS9/TCS9/ESP9 上或 ABS9/TCS9/ESP9 浸入水中，热的电控单元和连接器在降温过程中，可能会产生一个低压。这可能导致水通过线头如地线渗到连接器中。这意味着所有的导线，只要其线头所处位置潮湿或能被水溅到，就必须加以密封保护，以防止水分渗入线缆并通过线缆渗到连接器中。因此，对线头进行密封是必不可少的。对水能溅到的地方有线束接合处的情况同样需要采取密封措施。如果这些要求无法满足，系统可能由于 ECU 连接器腐蚀而失效。

It is possible that a lower pressure is created when the hot unit and connector is cooled down when the ABS9/TCS9/ESP9-unit is splashed by water or immersed into water. This can cause water to be soaked via the wire ends e.g. ground wires into the connector. This means that all wires, which end in a wet or splash water area must be protected that no water can penetrate into the wire and through the wire into the connector. The sealing of these wire ends is required. The same requirement is also valid for splices which are installed in splash water area. If this requirements cannot be fulfilled the system can fail due to corrosion in the ECU connector.

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接口部分验证 Interface Validation

ECU 连接器与线束连接器之间的接口需要由线束连接器供应商提供确认。博世可以提供壳体用于此验证试验。

The validation of the connector/header interface is the responsibility of the wiring harness connector supplier. Bosch supports this activity by providing housings for validation.

线束的防水密封性 Water tightness for the wiring harness

线束设计要满足的要求有很多，特别重要的是线束的密封性。水分可以通过虹吸作用、重力、破损或渗漏的线缆绝缘层以及压力差渗入线束中。对裸露的线头和线束接合处的最常用的一种密封方法是使用热缩管和密封胶，热缩管必须完全覆盖线头。如图1所示.

It is important that the following design requirements are met when designing a harness, especially regarding the tightness. Water can enter the wiring harness by capillary reaction, by gravity, damaged/porous wire insulation and by pressure differential. A way for sealing open wire ends and splices is to use shrink tube with sealing adhesive. The shrink tube must cover completely the wire end, see also Picture 1.

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图1：热缩管使用建议 Picture 1: Proposal of a shrink tube

密封的线束搭铁点和接合处必须通过在 0.5bar 压力下5分钟以上不泄漏的测试。线束供应商必须保证提供给客户的线束已经通过上面的线束搭铁点和接合处密封测试。同时，线束供应商还必须检查线束连接器的密封性，需要注意的是，38针脚连接器所使用的线径大小不是由连接器供应商规范规定的(在博世的电路图中有具体描述)。

Sealed wire eyelets and sealed splices must withstand at least a pressure leak test of 0.5bar (7psi) for at least 5min. The wiring harness supplier is responsible for testing all sealed wires and sealed splices before delivering the wiring harness to the customer. It is also within the wiring harness supplier's responsibility to check the tightness of the wiring harness connector, here 38-pin connector by using wire diameters which are not within the connector supplier specification.

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最佳布线实践: Best Practice for the Wiring Routing:

-由于水分因为重力作用可以通过线束传导，因此ECU要求安置在比线束中任何线头和接合处都高的位置；

-要采取措施防止线束周围有水分聚集。线束的塑料保护层要求能把水从线束边排走；

-布线必须绕开排水的区域；

-沿着锋利的边缘以及容易被石头撞击损害的线缆要加以保护；

-线束中进水会导致各信号线间相互干扰，如由于绝缘不好而导致的轮速传感器信号线间的 PWM 信号干扰；

-线束接合处和搭铁点必须安置在高压清洁器无法直接冲洗的地方；

-在容易溅到水的地方，线头必须布置得弯曲朝下。

- Due to the effects of gravity passing water through the wires the ECU should be packaged higher than any wire end or splice in the harness.

- Measures should be taken to prevent water from collecting around the wiring harness. Plastic wire covering/protection should allow water to drain from harness.

- Wiring should be routed away from water drainage areas.

- Wires that are routed around sharp edges or that can be damaged from stone impingements should be protected.

- Water in the wiring harness can cause interference on various signal wires, e.g. PWM-signal interference with wheel speed sensor wires due to poor wiring insulation.

- Wires splices and ground eyelets must be located away from direct contact with high pressure cleaner.

- Any wire end terminating in a splash water area should be orientated so that the wire end is pointing down.

连接器注意事项: Connector Precautions:

-在 OEM 封装过程中应要求不能有水，液体以及其他外部东西进入敞开的连接器 (公头或母头)；

-总装工厂必须保证连接器完全啮合，滑动锁止开关处于锁紧位置；

-接线端密封塞的尺寸和安装必须完全正确；

-所有的空腔中都应塞有密封插销或盲塞；

-接线端接插器必须满足加工的高度，宽度以及强度方面的要求。

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During the OEM assembly process care must be taken that no water/ fluid / foreign materials can enter an open connector (male or female).

The assembly plant must ensure that connector is fully engaged and slide or lever is in locked position.

Terminal wire seals must be of correct size and installed properly.

Seal plugs or posts must be installed in all open cavities.

Terminal wire crimps must meet manufactures height, width and strength requirements.

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20°C TCS作用时的马达电流 待定

Motor current during TCS - actuation at 20°C tbd

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20°C ABS制动时的马达电流 待定
条件：主缸压力200bar，车速100km/h

Motor current during ABS - braking at 20°C Condition: pressure at master cylinder = 200bar, vehicle speed = 100km/h tbd

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Wire / Conn.No	Wire		Designation	Current		max. permissible wire resistance R_L(mΩ)	min. leakage Resistance R_P(kΩ)
	from	to		max	min		
13	ground	K1/13	ground for pump motor	20..39A *	10A *	10	---
38	ground	K1/38	ground for solenoid valves and ECU	5…16A *	2.8A *	10	---
1	battery	K1/1	voltage supply for pump motor	20..39A *	10A *	10	200
25	battery	K1/25	voltage supply for solenoid valves	5…15A *	2A *	10	200
7	battery	K1/7	voltage supply for ECU	1A	800mA	60	200
35	WAU	K1/35	wake up function	100mA	50mA	150	200
32	K1/32	K35,K36/**	voltage supply for SAS and YRS (by use of separate YRS)	300mA	150mA	100	200
**,6,20, 31	wheel sp. connector	K1/**,6,20,31	signal wheel speed sensor FL,FR,RL,RR	16mA	6mA	250	200 to ground 1.5MΩ to battery
**,18,33, 19	wheel sp. connector	K1/**,18,33,19	voltage supply for active wheel speed sensor FL,FR,RL,RR	16mA	6mA	250	200 to ground 1.5MΩ to battery
**	other ECU	K1/**	wheel speed sensor output	20mA	10mA	250	200
**	other ECU	K1/**	vehicle speed output	50mA	30mA	250	200
**	EBD-warning lamp	K1/**	EBD warning lamp	30mA	5mA	250	200
**	ABS-warning lamp	K1/**	ABS warning lamp	30mA	5mA	250	200
**	MFL	K1/**	multifunctional lamp	50mA	30mA	250	200
**	ESP OFF	K1/**	ESP OFF	30mA	5mA	250	200
**	brake light switch	K1/**	brake light switch	10mA	5mA	250	200
**	brake switch	K1/**	brake switch	10mA	5mA	250	200
26	other ECU	K1/26	CAN1-H	30mA	20mA	250	200
14	other ECU	K1/14	CAN1-L	30mA	20mA	250	200
**	other ECU	K1/**	CAN2-H	30mA	20mA	250	200
**	other ECU	K1/**	CAN2-L	30mA	20mA	250	200
**	diagnostic connector	K1/**	diagnostic wire (only for application cars)	5mA	3mA	250	200
**	other ECU	K1/**	brake light actuation	200mA	150mA	100	200
**	other ECU	K1/**	handbrake switch	50mA	30mA	250	200
**	other ECU	K1/**	BBV (brake pad wear)	50mA	30mA	250	200
**	chassis	K35/**	ground for SAS	150mA	100mA	150	200
**	chassis	K35/**	ground for YRS (only by use separate YRS)	150mA	100mA	150	200

*只在 ABS 作用时 (ABS电流值高于 TCS 或 ESP, 马达电流最大值发生在 ABS 持续作用时) 。
*only during ABS actuation (the current values for ABS are higher than TCS or ESP, motor current value is worst case for steady state actuation).

**参考客户电路图。
**see customer specific circuit diagram.

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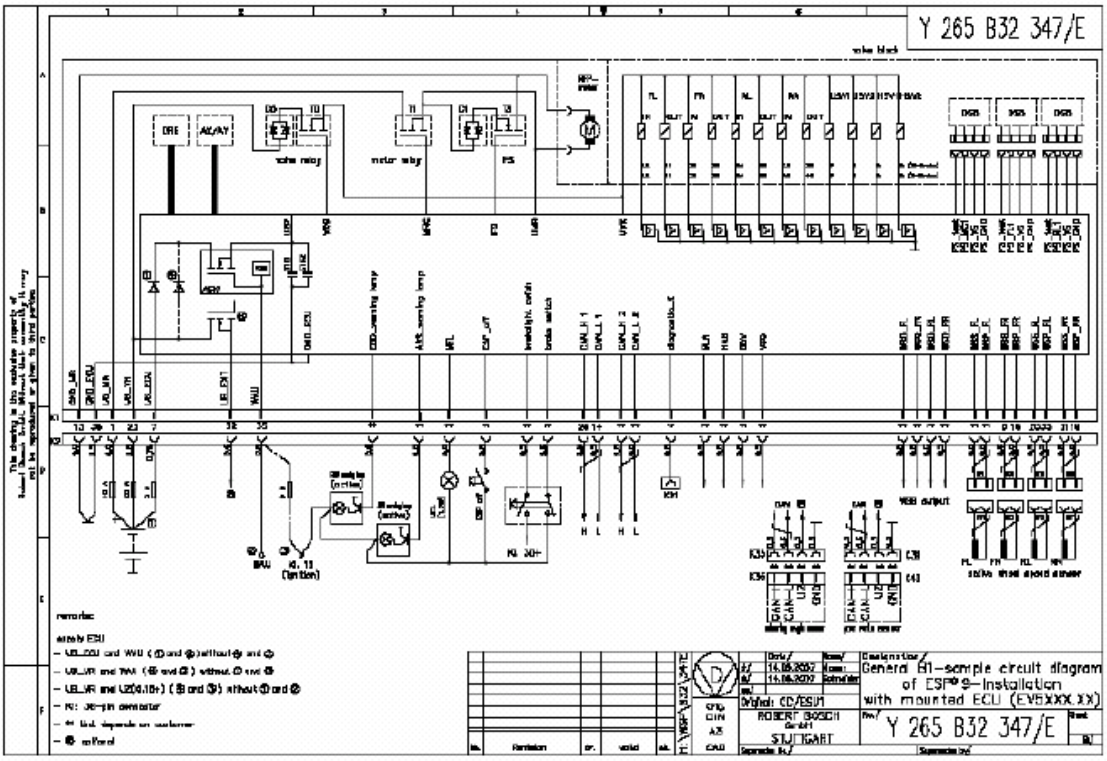
线束信息说明 Explanation for Wiring Harness Information

R_L为20°C时允许的最大电阻值。此值参考室温时的线缆阻值，额外的阻值如接触阻值在计算内部压降时也计入R_L。R_P为允许的各点对地或电源的泄漏阻值 (kΩ)。
R_L = Information of the maximum permissible wire resistance at 20°C.
This value refers to the electrical resistance of the wire at room temperature.
The additional resistance e.g. contact resistance is also calculated by R_L in the internal voltage drop calculation. R_P = value of permissible leakage per point in kΩ against ground or battery.



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